Property	Value	Reference
Molecular mass, g mol^{-1}	71.002	(Patnaik 2003)
Melting point, °C (K)	-206.8 (66.4)	(Patnaik 2003)
Boiling point, °C (K)	-128.75 (144.40)	(Patnaik 2003)
Liquid density at -128.75°C (144.40 K), kg m^{-3}	1.533	(Henderson and Woytek 2010)
Gas density at 101.3 kPa (1 atm) 21°C (294 K), kg m^{-3}	2.902	(Lide 2010)
Specific gravity at 101.3 kPa (1 atm) 21.1°C (294.2 K)	2.46 (air = 1)	(Technical Resources International 2001)
Heat of vaporization, kJ mol^{-1}	11.59	(Henderson and Woytek 2010)
Triple point, °C (K) 0.263 Pa	-206.8 (66.35)	(Henderson and Woytek 2010)
Critical temperature, °C (K)	-39.25 (233.90)	(Henderson and Woytek 2010)
Critical pressure, kPa (atm)	4530 (44.7)	(Henderson and Woytek 2010)
Critical volume, cm^3 mol^{-1} (m^3 mol^{-1})	123.8 (1.238×10^{-4})	(Henderson and Woytek 2010)
Heat of formation (ΔH_f), 25°C (298.15 K), 101.3 kPa, kJ mol^{-1}	-124.7	(Wagman et al. 1982)
Gibbs Free Energy of formation (ΔG_f), 25°C (298.15 K), 101.3 kPa, kJ mol^{-1}	-83.2	(Wagman et al. 1982)
Entropy (S), 25°C (298.15 K), 101.3 kPa, J mol^{-1}	260.73	(Wagman et al. 1982)
Heat capacity (C_p), 25°C (298.15 K), J mol^{-1} K^{-1}	53.1	(Wagman et al. 1982)
Water solubility, 101.3 kPa, 25°C (298.15 K), mol NF_3 mol^{-1} H_2O	1.4×10^{-5}	(Henderson and Woytek 2010)
Dipole moment, D	0.234	(Air Products 2010a)
N-F bond distance, nm	0.137	(Henderson and Woytek 2010)
F-N-F bond angle, °	102.1	(Henderson and Woytek 2010)
Strength of NF_2-F bond, kJ	14	(Kennedy and Colburn 1961)
Strength of NF-F bond, kJ	17	(Kennedy and Colburn 1961)
Strength of N-F bond, kJ	17	(Kennedy and Colburn 1961)